

Fine-Tuning Before/After Explorer

Fine-Tuning Local LLM for Drilling & Completions -- companion notebook

This notebook uses real exported answers from this book's own base model and Chapter 5 LoRA fine-tuned checkpoint. Its job is not just to show that fine-tuning changed answers, but to make the real pattern visible: the model learned many seen training examples, still failed the held-out report, and therefore later chapters need evaluation and retrieval.

Main takeaway

Out[1]=

Fine-tuning helped the model reproduce answers from seen training reports, but it did not give the model factual access to a report it never saw. That is why the book moves next from fine-tuning to retrieval.

1. Load the real exported data

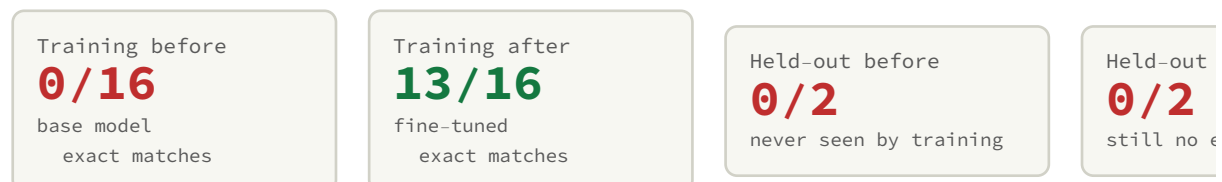
Out[7]= 18

Helper functions used by the visual sections below.

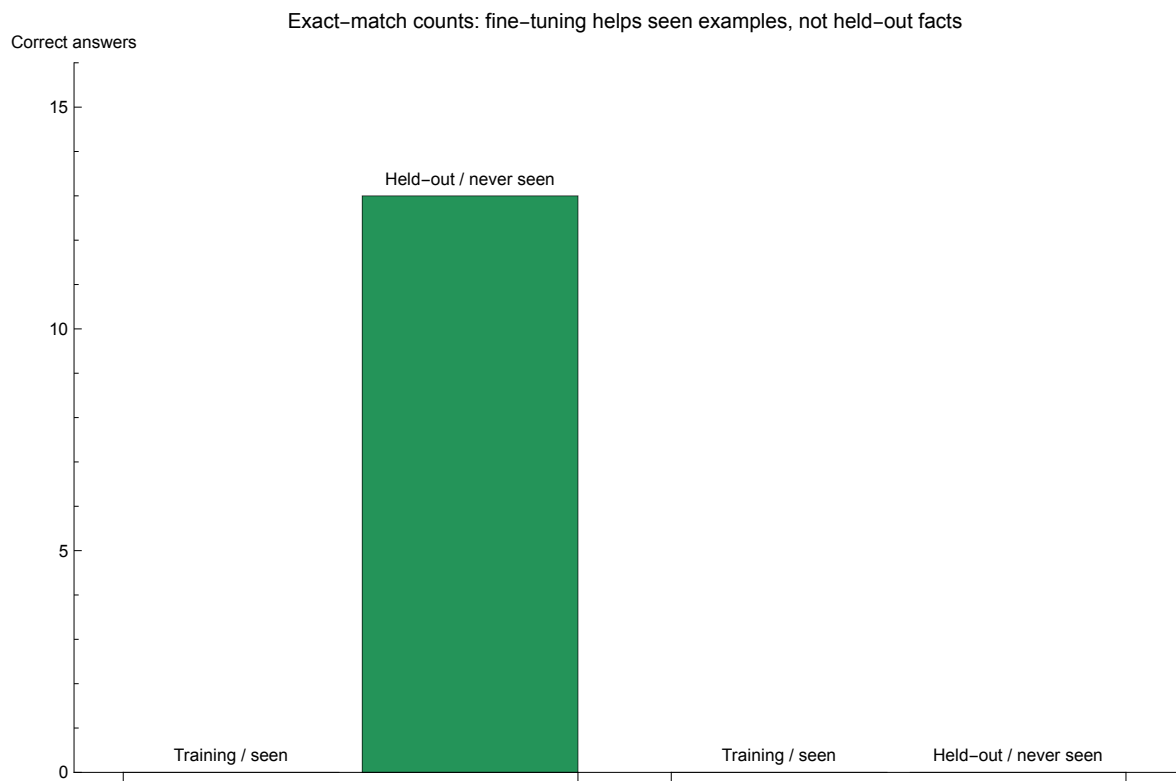
2. Executive view: what changed?

The headline is intentionally split into seen training questions and a never-seen held-out report. This is the key distinction: fine-tuning improved word overlap on examples like the ones it studied, but it did not create a grounded memory of the held-out report.

Out[36]=



Out[37]=



3. Outcome map: all questions at once

This compact map makes the pattern visible without clicking through examples one by one. Green means the answer exactly contained the reference wording. The word-overlap columns show how much of the correct answer's vocabulary appears in each model answer, which helps separate total misses from near misses.

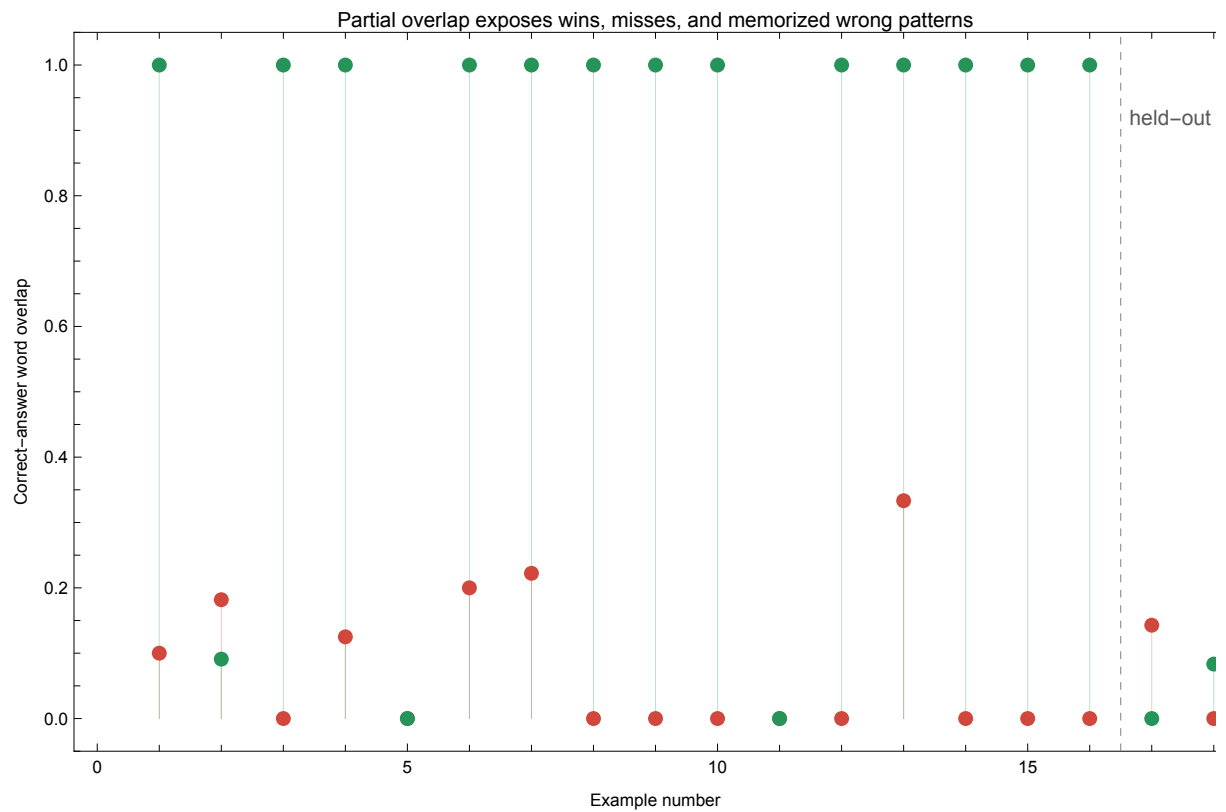
Out[38]=

Row	Set	Report	Question	Before	After	Before word overlap	After word overlap	Reading
#1	Training / seen	Report #3	Ops	MISS	MATCH	0.10	1.00	Exact match after fine- tuning
#2	Training / seen	Report #3	Plan	MISS	MISS	0.18	0.09	Partial wording overlap, wrong answer

#3	Training / seen	Report #19	Ops	MISS	MATCH	0.00	1.00	Exact match after fine- tuning
#4	Training / seen	Report #19	Plan	MISS	MATCH	0.13	1.00	Exact match after fine- tuning
#5	Training / seen	Report #36	Ops	MISS	MISS	0.00	0.00	Wrong memoriz- ed pattern
#6	Training / seen	Report #36	Plan	MISS	MATCH	0.20	1.00	Exact match after fine- tuning
#7	Training / seen	Report #38	Ops	MISS	MATCH	0.22	1.00	Exact match after fine- tuning
#8	Training / seen	Report #38	Plan	MISS	MATCH	0.00	1.00	Exact match after fine- tuning
#9	Training / seen	Report #39	Ops	MISS	MATCH	0.00	1.00	Exact match after fine- tuning
#10	Training / seen	Report #39	Plan	MISS	MATCH	0.00	1.00	Exact match after fine- tuning
#11	Training / seen	Report #48	Ops	MISS	MISS	0.00	0.00	Wrong memoriz- ed pattern
#12	Training / seen	Report #48	Plan	MISS	MATCH	0.00	1.00	Exact match after fine- tuning
#13	Training / seen	Report #49	Ops	MISS	MATCH	0.33	1.00	Exact match after fine- tuning

#14	Training / seen	Report #49	Plan	MISS	MATCH	0.00	1.00	Exact match after fine- tuning
#15	Training / seen	Report #50	Ops	MISS	MATCH	0.00	1.00	Exact match after fine- tuning
#16	Training / seen	Report #50	Plan	MISS	MATCH	0.00	1.00	Exact match after fine- tuning
#17	Held-out / never seen	Report #37	Ops	MISS	MISS	0.14	0.00	Held-out general: ization n gap
#18	Held-out / never seen	Report #37	Plan	MISS	MISS	0.00	0.08	Held-out general: ization n gap

Out[39]=



4. Query-to-answer cosine similarity

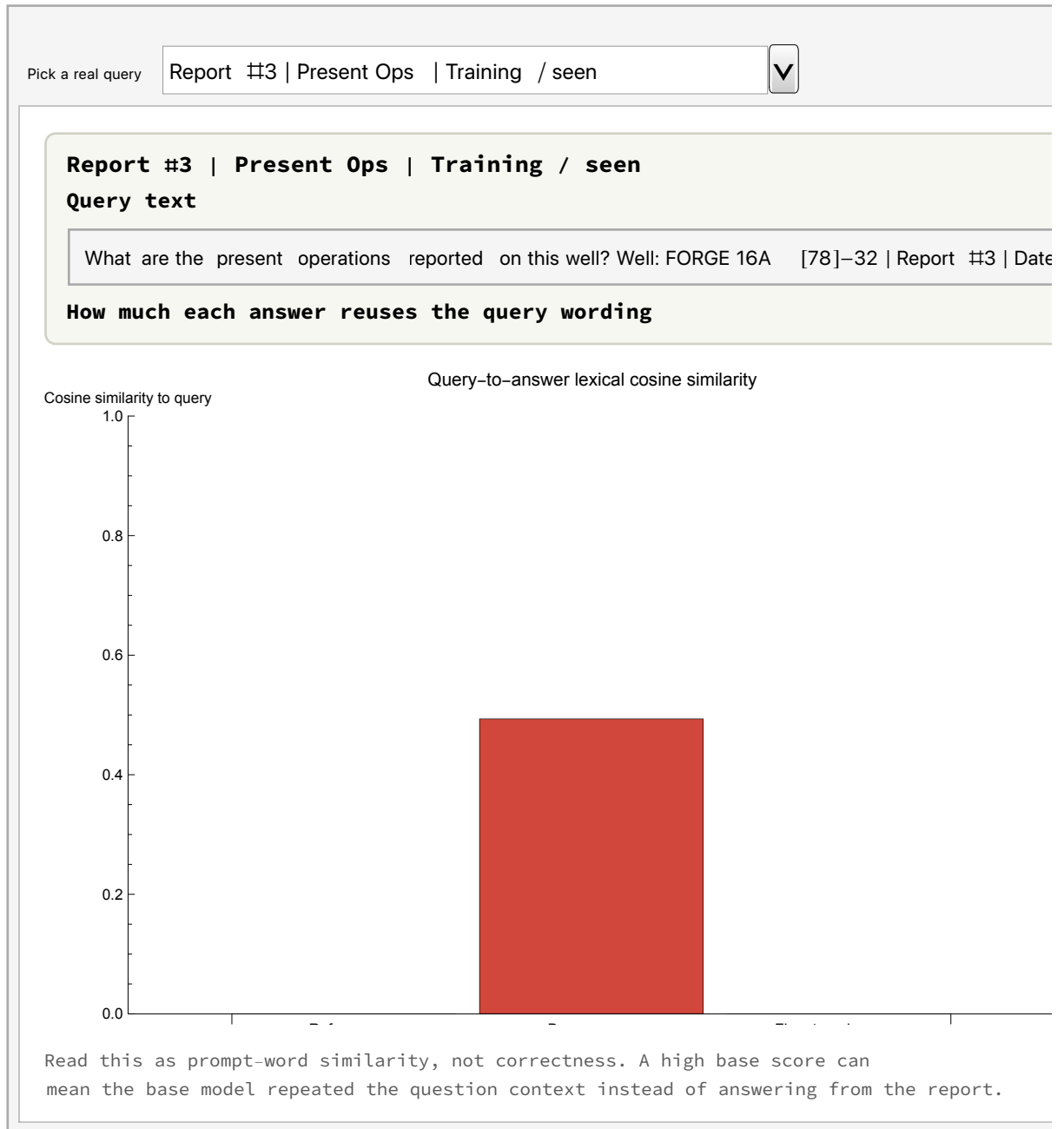
Cosine similarity compares the words in the query with the words in each answer. It is useful for spotting answers that echo the question, but it is not a correctness score. In this notebook, correctness still comes from matching the reference answer and reading the source excerpt.

Out[42]=

Row	Example	Query vs reference	Query vs base	Query vs fine-tuned	Reading
#1	Report #3 Present Ops Training / seen	0.00	0.49	0.00	Base echoes query more
#2	Report #3 Activity Planned Training / seen	0.00	0.16	0.00	Base echoes query more
#3	Report #19 Present Ops Training / seen	0.00	0.49	0.00	Base echoes query more
#4	Report #19 Activity Planned Training / seen	0.00	0.20	0.00	Base echoes query more
#5	Report #36 Present Ops Training / seen	0.00	0.60	0.00	Base echoes query more
#6	Report #36 Activity Planned Training / seen	0.00	0.31	0.00	Base echoes query more
#7	Report #38 Present Ops Training / seen	0.00	0.49	0.00	Base echoes query more
#8	Report #38 Activity Planned Training / seen	0.00	0.31	0.00	Base echoes query more
#9	Report #39 Present Ops Training / seen	0.00	0.49	0.00	Base echoes query more

#10	Report #39 Activity Planned Training / seen	0.00	0.31	0.00	Base echoes query more
#11	Report #48 Present Ops Training / seen	0.00	0.41	0.00	Base echoes query more
#12	Report #48 Activity Planned Training / seen	0.00	0.25	0.00	Base echoes query more
#13	Report #49 Present Ops Training / seen	0.00	0.41	0.00	Base echoes query more
#14	Report #49 Activity Planned Training / seen	0.00	0.25	0.00	Base echoes query more
#15	Report #50 Present Ops Training / seen	0.00	0.41	0.00	Base echoes query more
#16	Report #50 Activity Planned Training / seen	0.00	0.25	0.00	Base echoes query more
#17	Report #37 Present Ops Held-out / never seen	0.00	0.60	0.00	Base echoes query more
#18	Report #37 Activity Planned Held-out / never seen	0.00	0.31	0.00	Base echoes query more

Out[45]=



5. Guided examples: one win, one limit

These two examples are the shortest path through the result. The first shows what fine-tuning learned well. The second shows the limit: a held-out report still needs retrieval because its facts were never in the training examples.

Out[51]=

What fine-tuning learned

Report #3 | Present Ops | Training / seen

Question

What are the present operations reported on this well?

Expected

RIGGING UP WITH FRONTIER CREWS AND SUPERVISION AND JW JONES TRU

Before **MISS** 0.10 word overlap

Why base missed

The base prompt only includes the well name, report number, date, and quest

After **MATCH** 1.00 word overlap

After answer

RIGGING UP WITH FRONTIER CREWS AND SUPERVISION AND JW JONES TRU

Takeaway

The fine -tuned model reproduces the expected report wording exactly on a s

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Report #

Question

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6. Answer microscope

Pick any real question. Highlighted words are tokens that also appear in the reference answer. This turns the before/after comparison into something you can read visually: generic explanations, exact copied successes, and confident wrong template reuse look very different.

Out[54]=

Pick a real question 

Question What are the present operations reported on this well?

Context Well: FORGE 16A [78]-32 | Report #3 | Date: 2020-10-22

Set Training / seen

Outcome Before MISS After MATCH

Source report excerpt

PRESENT OPERATIONS:RIGGING UP WITH FRONTIER CREWS AND SUPERVISION AND JW JONES TRUCKING
ACTIVITY PLANNED:CONTINUE RIGGING UP
WITH FRONTIER CREWS AND SUPERVISION AND JW JONES TRUCKING

Reference answer

RIGGING UP WITH FRONTIER CREWS AND SUPERVISION AND JW JONES TRUCKING

Why the base model missed

The base prompt only includes the well name, report number, date, and question. It

Base model before fine-tuning	Fine-tuned model after fine-tuning
The report you've provided appears to for a well called "FORGE 16A and it is dated October 22, The specific details of what operations reported are not explicitly stated in message . However , based	RIGGING UP WITH FRONTIER CREWS AND AND JW JONES TRUCKING
Correct-answer word overlap: 0.10 Query cosine: 0.49	Correct-answer word overlap: 1.00 Query cosine: 0.00

7. Failure patterns

The failures after fine-tuning are the most important teaching examples. They show that the fine-tuned model can sound more domain-specific while still pulling the wrong remembered pattern into a new report context.

Out[58]=

Post -fine -tune failure Report #3 | Activity Planned | Training / seen | Partial wording overlap, wrong answer Report #36 | Present O

Partial wording overlap, wrong answer
 Report #3 | Activity Planned | Training / seen

Expected from report	CONTINUE RIGGING UP WITH FRONTIER CREWS AND SUPERVISION AN
Fine-tuned answer	CONTINUE BUILDING CURVE TO 65 °
What to notice	The answer is now domain-shaped, but it is the wrong rep

8. Why retrieval comes next

This Chapter 5 notebook deliberately stops at fine-tuning, but the visual result points toward Chapter 9: when the report was never in training, the model needs a retrieval step to find the relevant source text before answering.

Out[61]=

Held-out report #37 remains unsolved by fine-tuning alone

Fine-tuning changed the model's style and made many seen examples exact.

It did not make unseen report facts available inside the weights.

Next visual extension: add a third answer lane for Fine-tuned + retrieval, with the retrieved source chunk shown beside the generated answer.

Report	Question	Fine-tuned	Reference	Fine-tuned answer
Report #37	Present Ops	MISS	RIG UP SCHLUMBERGER PREPARE TO RUN LOGS	PIPE FREE, TRIP OUT OF HOLE FOR BHA INSPECTION
Report #37	Activity Planned	MISS	RUN UBI LOG, PICK UP BHA # 2, DRILL 8 3/4" Curve	CONTINUE BUILDING CURVE TO 65°

To regenerate this data yourself -- for example after retraining Chapter 5's checkpoint, or against your own reports -- run "python mathematica/export_before_after.py" from the repository root, with book/.venv active. See mathematica/README.md for details and prerequisites.